

Faizan Muhammad

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Google - DeepMind

Senior Software Engineer

July 2020 – Present

Mountain View, CA

- Search AI:
 - * Prototyping AI systems to retrieve, structure and utilize domain-specific knowledge and expertise
 - * Lead for vectorized chain-of-prompt framework powering LLM-based ranking heuristics and search experiences
 - * Developed production-grade tools for multi-step reasoning and knowledge representation
- Gemini:
 - * Productionized multi-modal Gemini models for serving, quantization, fine-tuning and evaluation [1]
 - * Translated experimental model code and checkpoints to the production-grade Gemini v1 launch
- TensorFlow:
 - * Re-architected tf.function by introducing new abstractions to radically improve consistency and performance
 - * *FunctionType*: comprehensive signature representation that controls type-checking, dispatch and tracing
 - * *AtomicFunction*: minimized Python overhead at the heart of tf.function and sped up Keras training by 4%

CTRL Labs (acquired by Meta)

Software Engineering Intern

May 2019 – August 2019

New York City, NY

- Engineered SLAM capabilities onto the wrist-mounted, EMG-based neural interface with a 3D camera attachment
- Devised interaction mechanisms combining localization and neural data and patented them [4]
- Orchestrated a neuromuscular teleoperation experience with a hexapod robot that was featured in an NPR show

Tufts University

Student, Research Assistant

Sep 2016 – May 2020

Boston, MA

- DARPA SAIL Open-World Novelty [2]
 - * Pioneered an AI agent architecture integrating symbolic and neural approaches to instill novelty resilience
 - * Led a group of graduate and undergraduate students to implement and publish the system architecture [3]
 - * AI Agent achieved top performance in independent external evaluations against other DARPA-selected teams
- ROS Augmented Reality Interface [3]
 - * A fast, high-bandwidth and accessible medium to convey robot perception, belief and planning in AR
 - * Won \$100K at Verizon 5G EdTech, received Computing Research Association award, and published it [3]
- Bachelor of Science in Computer Science
 - * *GPA*: 3.95, *Honors*: Summa Cum Laude, President @ Robotics Club
 - * *Highlight Courses*: RL, ML, Robotics, Computer Vision, Human Robot Interaction, Digital Circuits
 - * *Activities*: Actor/Writer/Director @ HYPE! Mime Troupe, President @ Robotics Club

International Mathematical Olympiad

Pakistani Team

Sep 2015 – Jul 2016

HKUST, Hong Kong

PUBLICATIONS

- [1] **Gemini: A Family of Highly Capable Multimodal Models** ArXiv 2023 (10839 Citations)
Faizan Muhammad ∈ Gemini Team Contributors (Page 63). [Paper Link](#)
- [2] **A Novelty-Centric Agent Architecture for Changing Worlds** AAMAS 2021 (35 Citations)
Faizan Muhammad, Vasanth Sarathy, Gyan Tatiya, Shivam Goel, Saurav Gyawali, Mateo Guaman, Jivko Sinapov, Matthias Scheutz. [Paper Link](#), [Presentation Link](#)
- [3] **Creating a Shared Reality with Robots** HRI 2019 (50 citations)
Faizan Muhammad, Amel Hassan, Andre Cleaver, Jivko Sinapov. [Paper Link](#), [Demo Link](#)
- [4] **Systems and methods for contextualized interactions with an environment** USPTO (14 Citations)
Faizan Muhammad, Daniel Wetmore. [Patent Link](#)